documentation (exercise 2)

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1 Experiment 1

1.1 Question 1

The Harris corner detector is theoretically rotation invariant because it classifies corners based on the eigenvalues of the second-moment matrix:

$$M = \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

A corner is detected when both eigenvalues are large. Under image rotation, intensity gradients transform consistently, preserving these eigenvalues and thus the Harris response:

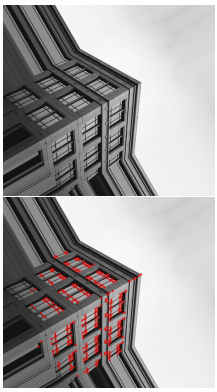
$$R = \det(M) - k \cdot (\text{trace}(M))^2$$

This response R is used to detect a corner: pixels with high R values that are local maxima in a neighborhood are classified as corners. This explains why, in theory, Harris corners are rotationally invariant.

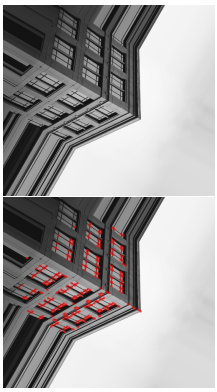
I performed preliminary tests by simply rotating the image and visualizing the detected corners. Visual inspection shows that the majority of corners remain consistent across rotations. This observation is further supported by quantitative analysis, as shown in Figure 2. Although the total number of detected keypoints varies slightly with rotation, most of them correspond closely to those in the original image. In fact, nearly all keypoints detected at different rotations can be matched to the original ones within a pixel-level tolerance. Specifically, we retain at least 97% of the keypoints, confirming the practical rotational robustness of the Harris detector. However, some small discrepancies still appear when the image is rotated. Although the Harris detector is theoretically rotation invariant, in practice interpolation during rotation slightly modifies the local gradients used to compute the second moment matrix. This can weaken certain corner responses below the detection threshold and introduce new responses in areas where intensity transitions become sharper. Additionally, the rotation introduces new image borders, which may artificially generate corners depending on the padding method. These implementation effects explain why a few keypoints may be lost or newly detected across rotations, despite the overall high correspondence.



(a) rotation: 0 degrees

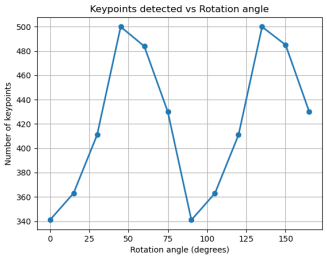


(b) rotation: 60 degrees

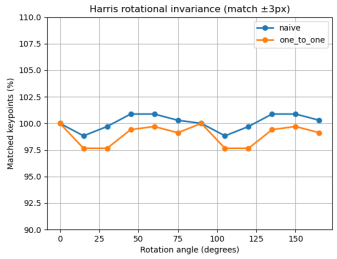


(c) rotation: 120 degrees

Figure 1: Visual comparison between keypoints vs the image rotation.



(a) number of keypoints detected



(b) matching points in the images

Figure 2: Quantitative data on rotational independency.

1.2 Question 2

The Harris corner detector is not scale-invariant because its corner response relies on a fixed-size window over image gradients. When the image is scaled up or down, the gradients no longer align with the same spatial support: a corner that produces a strong response R at one scale may become too small relative to the window (and disappear), or too large and resemble an edge instead of a corner. As a result, the same physical structure may only be detected as keypoint at a specific scale. A simple intuitive example is illustrated in Figure 3: a small L-shaped corner clearly produces a strong corner response at its native scale. However, when the same structure is observed at a much larger scale, the fixed detection window captures only gradual changes in one direction, leading the detector to classify it as an edge. Conversely, if the corner becomes too small when downscaled, it is smoothed away entirely and no longer detected. This

experiment clearly demonstrates how changes in scale alter corner detection and highlights the limitations of the Harris detector under scale transformations.

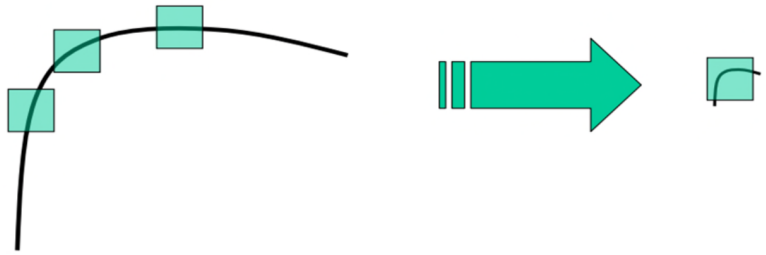
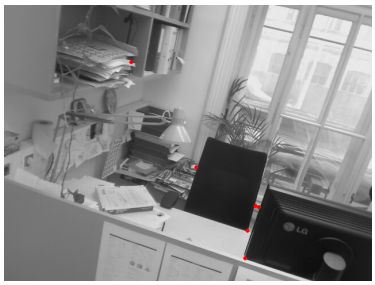


Figure 3: corners and edges over different scales.

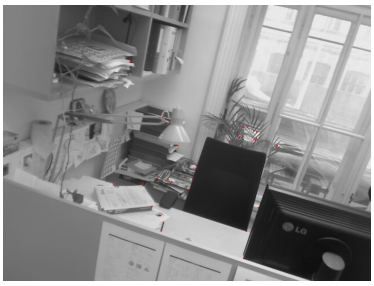
To visualize this limitation, I tested the detector on multiple scaled versions of the same image and compared the resulting keypoints. The visual results clearly show differences in the detected corners across scales. To quantify this, I computed the proportion of matching keypoints using the original image as reference and plotted a heatmap of the matches across scales. This reveals that the number of corresponding corners significantly decreases when the scale changes, confirming that Harris corners do not maintain consistency under zoom.



(a) Image 1: scale = 0.5



(b) Image 2: scale = 1.0



(c) Image 3: scale = 2.0

Figure 4: corner detector on different scale image

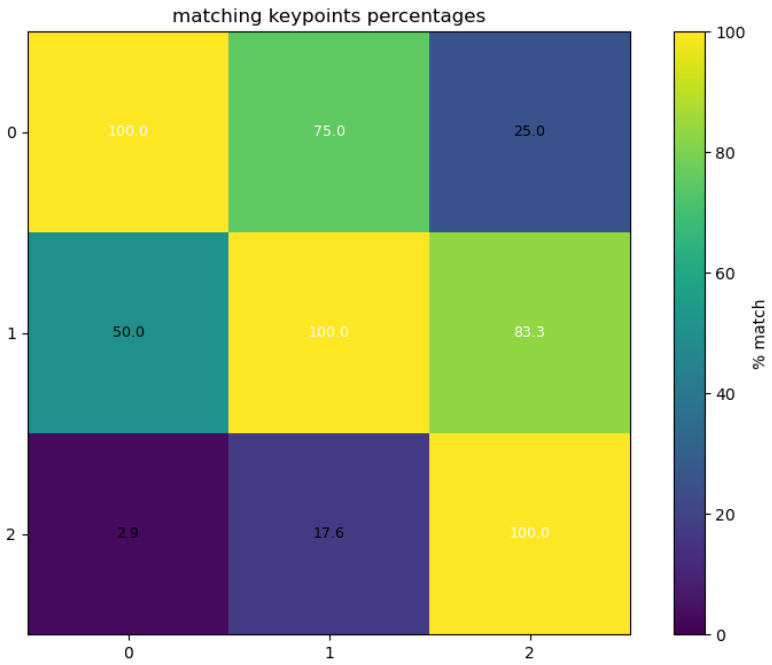


Figure 5: heatmap

2 Experiment 2

2.1 Question 1

Changing the patch size used to compute descriptors has a direct impact on the accuracy of keypoint matching. If the patch is very small, the descriptor contains only a limited amount of local image information, making different points appear similar and increasing the number of false matches. In contrast, a larger patch captures more context around the keypoint, which generally makes the descriptor more distinctive and improves matching accuracy. However, very large patches can introduce problems: they may include irrelevant background information or become sensitive to geometric distortions such as rotation and scale changes. This can cause descriptors of the same physical point to become dissimilar when transformations occur. Therefore, the patch size must be chosen as a trade-off: large enough to capture discriminative structure, but not so large that it includes unrelated or distorted content.

In the experiment I've tried different patch sizes and these are the results. At first we can see that overall we are able to identify good matches, but by investigating some quantitative values, we can clearly see the trade-off explained above.



Figure 6: Example of matches for different patch sizes.

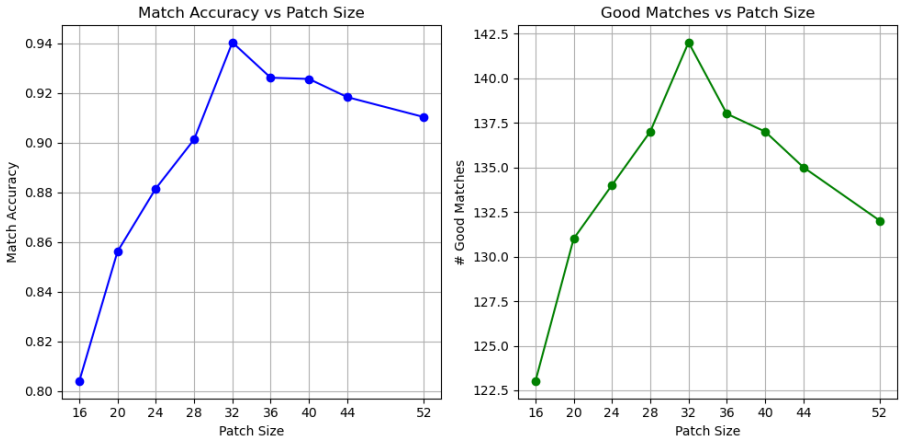
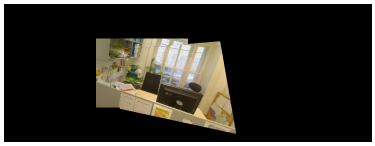


Figure 7: Statistics on the matches. Match accuracy:

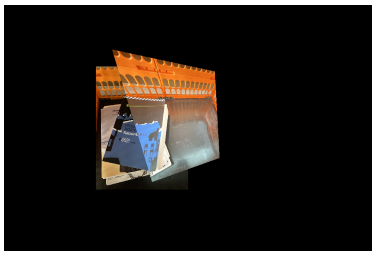
$$\text{match_accuracy} = \frac{\text{good_matches}}{\text{total_matches}}$$

2.2 Question 2

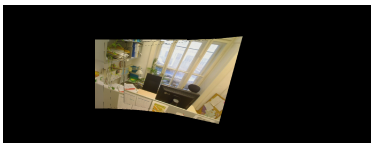
Image stitching performance is strongly influenced by both the properties of the images and the relative camera pose. In my experiments using **main_stitching.py** with the Harris corner detector, I observed that factors such as motion blur, uneven or varying illumination, and limited overlap between images can significantly degrade the results. Motion blur reduces the number of well-defined corners, making descriptor matching less reliable. Similarly, changes in lighting can alter intensity gradients, which Harris corners and their descriptors depend on, leading to fewer or mismatched keypoints. A small overlapping region between images also limits the number of correspondences, which can make homography estimation unstable or impossible.



(a) low number of correspondences



(b) different lights



(c) full stitching

Figure 8: stitching analysis

To mitigate these issues, descriptors can be improved in several ways. Increasing the patch size can make descriptors more robust to small local distortions or noise, while normalizing intensity values can reduce sensitivity to illumination changes.

3 Experiment 3

3.1 Question 1

Using least-squares to estimate the homography instead of RANSAC generally leads to significantly worse performance. This is because least-squares assumes that all matches are correct and does not handle matching errors: even a few outliers can heavily distort the homography and cause the object detection to fail completely. In contrast, RANSAC explicitly distinguishes between inliers and outliers and rejects incorrect correspondences, making the estimation much more robust.

Fine-tuning the Lowe ratio in the match filtering step can indeed reduce the number of outliers — a stricter threshold improves match reliability but may also reject valid correspondences, potentially resulting in an unstable or underdetermined homography. Therefore, adjusting the ratio only provides limited improvement, and a robust outlier-handling method like RANSAC remains essential.

In our experiments, I've observed that with appropriate Lowe ratio tuning the detector can correctly identify the object, whereas poor tuning leads to failures in matching and homography estimation. We also compared the simple least-squares approach with RANSAC and noticed a clear performance gap: in many parameter settings, least-squares deviates substantially and does not

produce a reliable alignment, confirming that robust estimation is crucial for object detection.

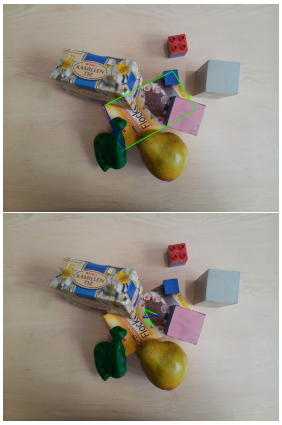


Figure 9: A good vs a bad result using LS

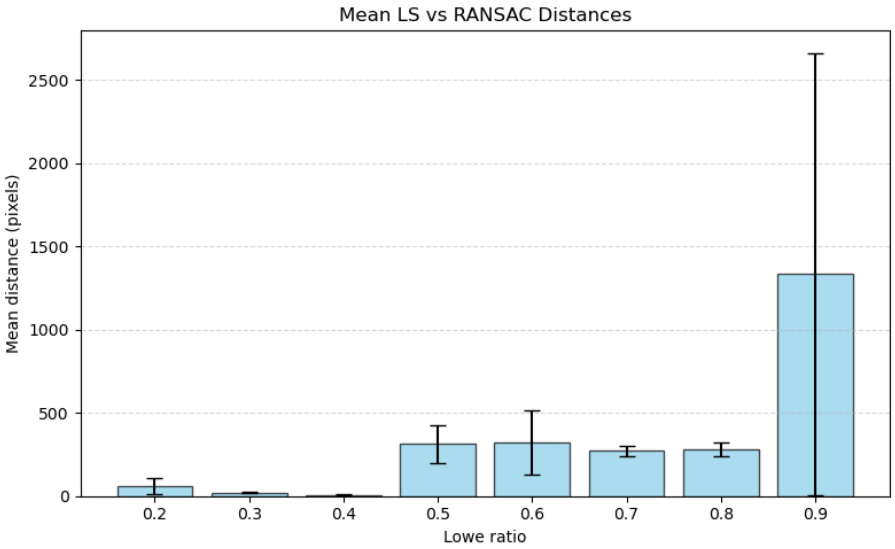


Figure 10: LS vs RANSAC distance

3.2 Question 2

Changing the RANSAC parameters has a strong influence on both robustness and computation time. Increasing the confidence parameter makes RANSAC more likely to find a correct homography, but it also requires more iterations,

increasing runtime. The inlier threshold affects how strictly RANSAC evaluates correspondences: a smaller threshold is more selective and usually produces a more accurate homography, but may classify many valid matches as outliers and lead to fewer inliers or even failure. Conversely, a larger threshold accepts more matches as inliers, reducing the number of required iterations, but risks including incorrect correspondences and decreasing accuracy. Therefore, a good balance between confidence and inlier threshold is required to achieve both reliable and efficient homography estimation.

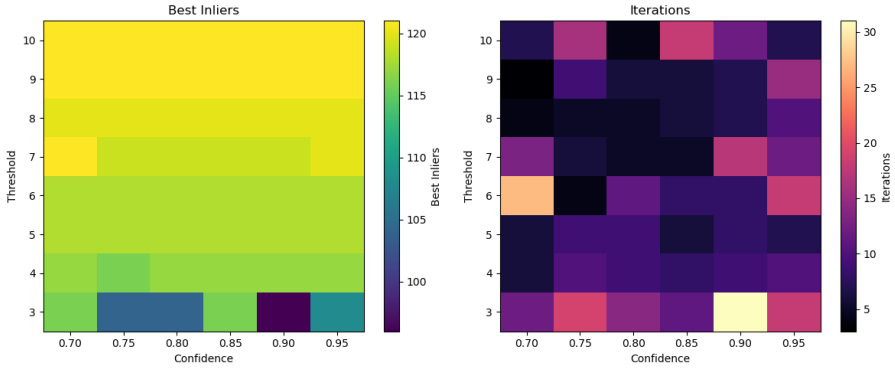


Figure 11: Inliers and iterations.

Here we can note that the confidence doesn't change that much the number of inliers found with respect to the threshold. This allows us to conclude that, for the images used, the homography is found reliably even with relatively few iterations. Increasing the confidence mostly affects the number of RANSAC iterations, but does not improve the inlier set when a correct homography is already found early. In contrast, the inlier threshold directly controls which matches are accepted as geometrically consistent, making it the parameter with the most visible influence on performance.